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Class :- SYIT	Batch :- B1
Date of Assignment :- 25-08-2026	Date of Submission :- 25-08-2026

Practical 21

Code -

```
ThisBuild / scalaVersion := "2.13.12"

libraryDependencies += Seq(
  "org.apache.spark" %% "spark-core" % "3.5.1",
  "org.apache.spark" %% "spark-sql" % "3.5.1"
)

lazy val root = (project in file("."))
  .settings(
    name := "Jasad1"
  )
```

```
import org.apache.spark.sql.SparkSession

object Jasad extends App {
  val s = SparkSession.builder()
    .appName("JoinTwoCSV")
    .master("local[*]")
    .getOrCreate()
  val d1 = s.read
    .option("header", "true")
    .csv("file1.csv")
  val d2 = s.read
    .option("header", "true")
    .csv("file2.csv")
  val j = d1.join(d2, Seq("id"), "inner")
  j.show()
  j.write.option("header", "true")
  s.stop()
}
```

	build.sbt	Jasad.scala	file1.csv ×	file2.csv
1		id,name		
2		1,rohit		
3		2,priya		

	build.sbt	Jasad.scala	file1.csv	file2.csv ×
1		id,score		
2		2,90		
3		1,70		

Output -

```

Run  Jasad ×
20/08/18 11:55:19 INFO DAGScheduler: JOB 3 FINISHED. SHOW at Jasad.scala:13, took 0.214514 s
26/08/18 11:55:20 INFO CodeGenerator: Code generated in 31.7345 ms
+---+-----+-----+
| id| name|score|
+---+-----+-----+
| 1|rohit| 70|
| 2|priya| 90|
+---+-----+-----+

26/08/18 11:55:20 INFO SparkContext: SparkContext is stopping with exitCode 0.
26/08/18 11:55:20 INFO SparkUI: Stopped Spark web UI at http://DESKTOP-2JJMAUE:4040
26/08/18 11:55:20 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/08/18 11:55:20 INFO MemoryStore: MemoryStore cleared
26/08/18 11:55:20 INFO BlockManager: BlockManager stopped
26/08/18 11:55:20 INFO BlockManagerMaster: BlockManagerMaster stopped
26/08/18 11:55:20 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator

```

Practical 22

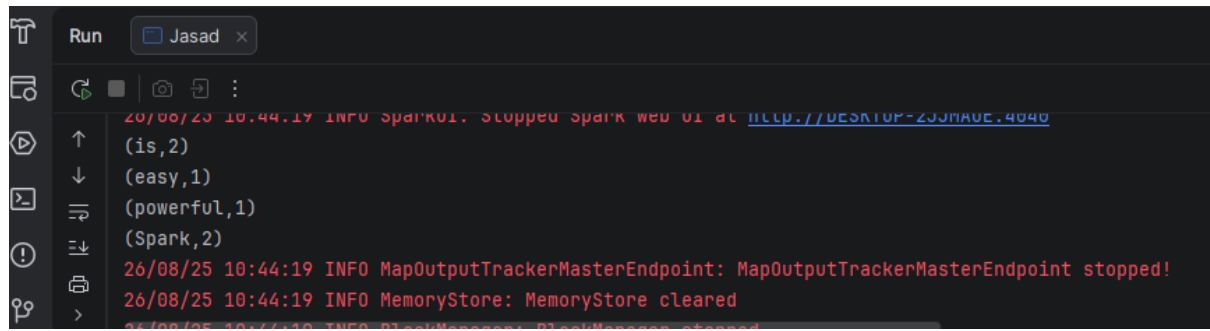
Code -

```
build.sbt x  input.txt  Jasad.scala
1 ThisBuild / scalaVersion := "2.12.18"
2
3 lazy val root = (project in file("."))
4   .settings(
5     name := "Jasad3"
6   )
7 ThisBuild / scalaVersion := "2.12.18"
8
9 libraryDependencies += Seq(
10   "org.apache.spark" %% "spark-core" % "3.5.1",
11   "org.apache.spark" %% "spark-sql" % "3.5.1"
12 )
```

```
build.sbt  input.txt x  Jasad.scala
1 Spark is easy
2 Spark is powerful
```

```
build.sbt  input.txt  Jasad.scala x
1 import org.apache.spark.sql.Session
2 object Jasad extends App {
3   val s = Session.builder()
4     .appName("WordCountExample")
5     .master("local[*]")
6     .getOrCreate()
7   val r = s.sparkContext.textFile("input.txt")
8   val w = r.flatMap(line => line.split("\\s+"))
9   val wo = w.map(word => (word, 1))
10  val wc = wo.reduceByKey(_ + _)
11  wc.collect().foreach(println)
12  s.stop()
13 }
```

Output -



The screenshot shows the 'Run' console of a Jupyter Notebook. The console title is 'Jasad'. The output consists of several lines of text. The first line is a log message: '26/08/25 10:44:19 INFO SparkUI: Stopped Spark web UI at http://DESKTOP-2J3NA0E:4040'. This is followed by four lines of data output: '(is,2)', '(easy,1)', '(powerful,1)', and '(Spark,2)'. Below the data output are three more log messages: '26/08/25 10:44:19 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!', '26/08/25 10:44:19 INFO MemoryStore: MemoryStore cleared', and '26/08/25 10:44:19 INFO BlockManager: BlockManager stopped'. The console also features a vertical toolbar on the left with icons for running, saving, and other notebook functions.

```
26/08/25 10:44:19 INFO SparkUI: Stopped Spark web UI at http://DESKTOP-2J3NA0E:4040
(is,2)
(easy,1)
(powerful,1)
(Spark,2)
26/08/25 10:44:19 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/08/25 10:44:19 INFO MemoryStore: MemoryStore cleared
26/08/25 10:44:19 INFO BlockManager: BlockManager stopped
```

Practical 23

Code -

```
1 ThisBuild / scalaVersion := "2.12.18"
2
3 libraryDependencies += Seq(
4   "org.apache.spark" %% "spark-core" % "3.5.1",
5   "org.apache.spark" %% "spark-sql" % "3.5.1"
6 )
7 lazy val root = (project in file("."))
8   .settings(
9     name := "Jasad5"
10  )
```

```
build.sbt  superstore.csv  Jasad.scala
1 Row ID,Order ID,Order Date,Ship Date,Ship Mode,Customer ID,Customer Name,Segment,Co 2342 ^ v:
2 1,CA-2016-152156,11/8/2016,11/11/2016,Second Class,C6-12520,Claire Gute,Consumer,United States,H
3 2,CA-2016-152156,11/8/2016,11/11/2016,Second Class,C6-12520,Claire Gute,Consumer,Unit States,H
4 3,CA-2016-138688,6/12/2016,6/16/2016,Second Class,DV-13045,Darrin Van Huff,Corporate,United Stat
5 4,US-2015-108966,10/11/2015,10/18/2015,Standard Class,SO-20335,Sean O'Donnell,Consumer,United St
6 5,US-2015-108966,10/11/2015,10/18/2015,Standard Class,SO-20335,Sean O'Donnell,Consumer,United St
7 6,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Stat
8 7,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Stat
9 8,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Stat
10 9,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Stat
11 10,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Sta
12 11,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Sta
13 12,CA-2014-115812,6/9/2014,6/14/2014,Standard Class,BH-11710,Brosina Hoffman,Consumer,United Sta
14 13,CA-2017-114412,4/15/2017,4/20/2017,Standard Class,AA-10480,Andrew Allen,Consumer,United State
15 14,CA-2016-161389,12/5/2016,12/10/2016,Standard Class,IM-15070,Irene Maddox,Consumer,United Stat
16 15,US-2015-118983,11/22/2015,11/26/2015,Standard Class,HP-14815,Harold Pawlan,Home Office,United
17 16,US-2015-118983,11/22/2015,11/26/2015,Standard Class,HP-14815,Harold Pawlan,Home Office,United
18 17,CA-2014-105893,11/11/2014,11/18/2014,Standard Class,PK-19075,Pete Kriz,Consumer,United States
19 18,CA-2014-167164,5/13/2014,5/15/2014,Second Class,AG-10270,Alejandro Grove,Consumer,United Stat
20 19,CA-2014-143336,8/27/2014,9/1/2014,Second Class,ZD-21925,Zuschuss Donatelli,Consumer,United St
21 20,CA-2014-143336,8/27/2014,9/1/2014,Second Class,ZD-21925,Zuschuss Donatelli,Consumer,United St
22 21,CA-2014-143336,8/27/2014,9/1/2014,Second Class,ZD-21925,Zuschuss Donatelli,Consumer,United St
23 22,CA-2016-137330,12/9/2016,12/13/2016,Standard Class,KB-16585,Ken Black,Corporate,United States
24 23,CA-2016-137330,12/9/2016,12/13/2016,Standard Class,KB-16585,Ken Black,Corporate,United States
25 24,US-2017-156909,7/16/2017,7/18/2017,Second Class,SF-20065,Sandra Flanagan,Consumer,United Stat
```

```
import org.apache.spark.sql.SparkSession
object Jasad extends App {
  val spark = SparkSession.builder()
    .appName("FilterCSVExample")
    .master("local[*]")
    .getOrCreate()
  import spark.implicits._
  val df = spark.read
    .option("header", "true")
    .option("inferSchema", "true")
    .csv("src/main/resources/superstore.csv")
  val f = df.filter($"Sales" > 5000)
  println("Filtered Data:")
  f.show()
  spark.stop()
}
```

Output -

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Run Alt+4

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26/08/25 11:53:48 INFO CodeGenerator: Code generated in 22.3827 ms

26/08/25 11:53:48 INFO SparkContext: SparkContext is stopping with exitCode 0.

ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region
	2698 CA-2014-145317	3/18/2014 3/23/2014	Standard Class	SM-20320	Sean Miller Home Office	United States	Jacksonville	Florida	32216	South TE		
	6827 CA-2016-118689	10/2/2016 10/9/2016	Standard Class	TC-20980	Tamara Chand	Corporate	United States	Lafayette	Indiana	47905	Central TE	

26/08/25 11:53:48 INFO SparkUI: Stopped Spark web UI at <http://DESKTOP-2JJMAVE:4040>

26/08/25 11:53:48 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!

26/08/25 11:53:48 INFO MemoryStore: MemoryStore cleared

26/08/25 11:53:48 INFO BlockManager: BlockManager stopped

26/08/25 11:53:48 INFO BlockManagerMaster: BlockManagerMaster stopped

26/08/25 11:53:48 INFO OutputCommitCoordinator\$OutputCommitCoordinatorEndpoint: OutputCommitCoordinator stopped!

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Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
TEC-MA-10002412	Technology	Machines	Cisco TelePresenc...	22638.48	6	0.5	-1811.0784
TEC-CO-10004722	Technology	Copiers	Canon imageCLASS ...	17499.95	5	0	8399.976

Practical 24

Code -

```
build.sbt  superstore.csv  Jasad.scala x
1  import org.apache.spark.sql.SparkSession
2  import org.apache.spark.sql.functions._
3  object Jasad extends App {
4      val spark = SparkSession.builder()
5          .appName("AverageSales")
6          .master("local[*]")
7          .getOrCreate()
8      val df = spark.read
9          .option("header", "true")
10         .option("inferSchema", "true")
11         .csv("src/main/resources/superstore.csv")
12     val result = df.groupBy("Category")
13         .agg(avg("Sales").alias("Average Sales"))
14     println("Average Sales by Category:")
15     result.show()
16     spark.stop()
17 }
18
```

Output -

```
26/08/25 12:22:32 INFO TaskSchedulerImpl: Killing all running tasks in stage 4: Stage finished
26/08/25 12:22:32 INFO DAGScheduler: Job 3 finished: show at Jasad.scala:15, took 0.083467 s
26/08/25 12:22:32 INFO CodeGenerator: Code generated in 6.433 ms
26/08/25 12:22:32 INFO SparkContext: SparkContext is stopping with exitCode 0.

+-----+
|      Category|      Average Sales|
+-----+
|Office Supplies|121.69225531914947|
|      Furniture| 353.4459311957568|
|      Technology| 454.5405475802047|
+-----+

26/08/25 12:22:32 INFO SparkUI: Stopped Spark web UI at http://DESKTOP-2JJMAUE:4040
26/08/25 12:22:32 INFO MapOutputTrackerMasterEndpoint: MapOutputTrackerMasterEndpoint stopped!
26/08/25 12:22:32 INFO MemoryStore: MemoryStore cleared
```